



COMP – 304

Summer 2026



Introduction to Kotlin

Objectives:

- ☐ Review the basics of Kotlin programming language:
 - Variables
 - Control structures
 - Functions
 - Classes
 - Collections
- ☐ Set up Android Studio to use Kotlin and build apps
- ☐ Develop Android Apps Using Kotlin



What is Kotlin

- ❑ Statically typed programming language for modern multiplatform applications - **type** checking is done at compile-time
- ❑ 100% interoperable with Java and Android

What does it look like?

Concise, simple and very easy to read (and write)

```
package hello
```

Optional package header

```
fun main(args: Array<String>) {  
    println("Hello World!")  
}
```

Package-level function, which takes an Array of strings as a parameter

Have you noticed?
Semicolons are optional



Basic Syntax

- ❑ Package specification should be at the top of the source file:

```
package my.demo
```

```
import java.util.*
```

```
// ...
```

- ❑ It is not required to match directories and packages: source files can be placed arbitrarily in the file system.



Defining variables

❑ Basic types include: Int, Long, Boolean, Double, and Char.

❑ Assign-once (**read-only**) local variable:

```
val a: Int = 1 // immediate assignment
```

```
val b = 2 // `Int` type is inferred
```

```
val c: Int // Type required when no initializer is provided
```

```
c = 3 // deferred assignment
```

❑ **Mutable** variable:

```
var x = 5 // `Int` type is inferred
```

```
x += 1
```

❑ **Top-level** variables:

```
val PI = 3.14
```

```
var x = 0
```

```
fun incrementX() {
```

```
    x += 1
```

```
}
```



Defining Functions

- ❑ Function having two `Int` parameters with `Int` return type:

```
fun sum(a: Int, b: Int): Int {  
    return a + b  
}
```

- ❑ Function with an expression body and inferred return type:

```
fun sum(a: Int, b: Int) = a + b
```

- ❑ Function returning no meaningful value:

```
fun printSum(a: Int, b: Int): Unit {  
    println("sum of $a and $b is ${a + b}")  
}
```

- ❑ `Unit` return type can be omitted:

```
fun printSum(a: Int, b: Int) {  
    println("sum of $a and $b is ${a + b}")  
}
```



Comments

- ❑ Just like Java and JavaScript, Kotlin supports end-of-line and block comments.

`// This is an end-of-line comment`

`/* This is a block comment
on multiple lines. */`

- ❑ Unlike Java, block comments in Kotlin can be nested.



Using string templates

```
var a = 1
```

```
// simple name in template:
```

```
val s1 = "a is $a"
```

```
a = 2
```

```
// arbitrary expression in template:
```

```
val s2 = "${s1.replace("is", "was")}, but now is $a"
```



Using conditional expressions

```
fun maxOf(a: Int, b: Int): Int {  
    if (a > b) {  
        return a  
    } else {  
        return b  
    }  
}
```

❑ Using if as an expression:

```
fun maxOf(a: Int, b: Int) = if (a > b) a else b
```



Using nullable values and checking for null

- ❑ Nullable types are denoted with a question mark operator.
- ❑ Return `null` if `str` does not hold an integer:

```
fun parseInt(str: String): Int? {  
    // ...  
}
```

- ❑ Use a function returning nullable value:

```
fun printProduct(arg1: String, arg2: String) {  
    val x = parseInt(arg1)  
    val y = parseInt(arg2)  
    // Using `x * y` yields error because they may hold nulls.  
    if (x != null && y != null) {  
        // x and y are automatically cast to non-nullable after null check  
        println(x * y)  
    }  
    else {  
        println("either '$arg1' or '$arg2' is not a number")  
    }  
}
```



Using type checks and automatic casts

- ❑ The `is` operator checks if an expression is an instance of a type.
 - If an immutable local variable or property is checked for a specific type, there's no need to cast it explicitly:

```
fun getStringLength(obj: Any): Int? {  
    if (obj is String) {  
        // `obj` is automatically cast to `String` in this branch  
        return obj.length  
    }  
  
    // `obj` is still of type `Any` outside of the type-checked branch  
    return null  
}
```



Using a for loop

```
val items = listOf("apple", "banana", "kiwi")  
for (item in items) {  
    println(item)  
}
```

or

```
val items = listOf("apple", "banana", "kiwi")  
for (index in items.indices) {  
    println("item at $index is ${items[index]}")  
}
```



Using a while loop

```
val items = listOf("apple", "banana", "kiwi")  
var index = 0  
while (index < items.size) {  
    println("item at $index is ${items[index]}")  
    index++  
}
```



Using **when** expression

```
fun describe(obj: Any): String =  
when (obj) {  
    1      -> "One"  
    "Hello" -> "Greeting"  
    is Long  -> "Long"  
    !is String -> "Not a string"  
    else     -> "Unknown"  
}
```



Using ranges

- ❑ Check if a number is within a range using in operator:

```
val x = 10
val y = 9
if (x in 1..y+1) {
    println("fits in range")
}
```

- ❑ Check if a number is out of range:

```
val list = listOf("a", "b", "c")

if (-1 !in 0..list.lastIndex) {
    println("-1 is out of range")
}

if (list.size !in list.indices) {
    println("list size is out of valid list indices range too")
}
```



Iterating over a range

```
for (x in 1..5) {  
    print(x)  
}
```

or over a progression:

```
for (x in 1..10 step 2) {  
    print(x)  
}  
println()  
for (x in 9 downTo 0 step 3) {  
    print(x)  
}
```



Using collections

- ❑ Iterating over a collection:

```
for (item in items) {  
    println(item)  
}
```

- ❑ Checking if a collection contains an object using in operator:

```
when {  
    "orange" in items -> println("juicy")  
    "apple" in items -> println("apple is fine too")  
}
```



Using Lambda Expressions

- ❑ Using lambda expressions to filter and map collections:

fruits

```
.filter { it.startsWith("a") }  
.sortedBy { it }  
.map { it.toUpperCase() }  
.forEach { println(it) }
```

- ❑ Creating basic classes and their instances:

```
val rectangle = Rectangle(5.0, 2.0) //no 'new' keyword required  
val triangle = Triangle(3.0, 4.0, 5.0)
```



Android Example - KotlinLanguageBasics

```
1 package com.example.kotlinlanguagebasics;
2 //
3 import androidx.appcompat.app.AppCompatActivity
4 import androidx.compose.foundation.layout.Column
5 import androidx.compose.foundation.layout.Spacer
6 import androidx.compose.foundation.layout.height
7 import androidx.compose.material3.Text
8 import androidx.compose.runtime.remember
9 import androidx.compose.ui.Modifier
10 import androidx.compose.ui.text.font.FontWeight
11 import androidx.compose.ui.unit.dp
12
13 class MainActivity : AppCompatActivity() {
14     override fun onCreate(savedInstanceState: Bundle?) {
15         super.onCreate(savedInstanceState)
16         setContent {
17             MyApp()
18         }
19     }
20
21     @Composable
22     fun MyApp() {
23         Column { this: ColumnScope
24             val (resultText, setResultText) = remember { mutableStateOf("") }
25
26             Text(text = resultText, modifier = Modifier.height(40.dp),
27                 fontWeight = FontWeight.Bold)
28
29             Spacer(modifier = Modifier.height(16.dp)) // Add spacing between elements
30
31             ActionBar(text = "Functions") {
32                 val sum = sum(a: 5, b: 6)
33                 setResultText("sum=$sum")
34                 printSum1(a: 7, b: 8)
35             }
36         }
37     }
38 }
```

Running Devices: Pixel 7a API 34

Live Edit disabled

sum=11

Functions

Loops

When

Lambda

if Keyword

Classes

Data Classes

Run app

2024-04-18 15:12:54: Launching app on 'Pixel 7a API 34'.

Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.example.kotlinlanguagebasics/.MainActivity }

KotlinLanguageBasics > app > src > main > java > com > example > kotlinlanguagebasics > MainActivity.kt

2:3 LF UTF-8 4 spaces



Android Example - KotlinLanguageBasics

```
class MainActivity : ComponentActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContent {  
            MyApp()  
        }  
    }  
  
    @Composable  
    fun MyApp() {  
        Column {  
            val (resultText, setResultText) = remember { mutableStateOf("") }  
  
            Text(text = resultText, modifier = Modifier.height(30.dp)) // Display  
area for results
```



Android Example - KotlinLanguageBasics

Spacer(modifier = Modifier.height(16.dp)) // Adds space between the text and buttons

```
ActionButton("Functions") {  
    val sum = sum(5, 6)  
    setResultText("sum=$sum")  
    printSum1(7, 8)  
}  
ActionButton("Loops") {  
    val numbers = listOf(1, 2, 3, 4, 5, 6)  
    setResultText("numbers: ${numbers.joinToString(", ")}")  
}  
ActionButton("When") {  
    val str = describe("some string")  
    setResultText("description=$str")  
}
```



Android Example - KotlinLanguageBasics

```
ActionButton("Lambda") {  
    val sumLambda = { a: Int, b: Int -> a + b }  
    val result = sumLambda(9, 3)  
    setResultText("sum from lambda=$result")  
}  
ActionButton("It Keyword") {  
    val numbers = arrayOf(1, -2, 3, -4, 5)  
    val listOfPositiveNumbers = numbers.filter { it > 0 }  
    setResultText("positive numbers:  
${listOfPositiveNumbers.joinToString(", ")}")  
}  
ActionButton("Classes") {  
    val person = Person("John", "Stone", 27)  
    setResultText("person = ${person.upperCaseFirstName},  
${person.upperCaseLastName}, ${person.age}")  
}
```



Android Example - KotlinLanguageBasics

```
ActionButton("Data Classes") {  
    val user = User("john stone", 27)  
    setResultText("user = ${user.name}, ${user.age}")  
}  
}  
}
```

@Composable

```
fun ActionButton(text: String, action: () -> Unit) {  
    Button(onClick = action) {  
        Text(text = text)  
    }  
}
```



Android Example - KotlinLanguageBasics

// Helper functions

```
private fun describe(obj: Any): String =  
    when (obj) {  
        1 -> "One"  
        "Hello" -> "Greeting"  
        is Long -> "Long"  
        !is String -> "Not a string"  
        else -> "Unknown"  
    }
```

```
private fun sum(a: Int, b: Int): Int = a + b
```

```
private fun printSum1(a: Int, b: Int) {
```

```
    // Normally you'd update a state to show a toast, omitted for brevity
```

```
}
```



Android Example - KotlinLanguageBasics

```
data class User(val name: String, val age: Int)
```

```
class Person(firstName: String, lastName: String, personAge: Int) {  
    val upperCaseFirstName = firstName.uppercase()  
    val upperCaseLastName = lastName.uppercase()  
    var age = personAge  
}
```



References

- ❑ <https://codelabs.developers.google.com/codelabs/build-your-first-android-app-kotlin/index.html#0>
- ❑ <https://kotlinlang.org/>
- ❑ <https://kotlinlang.org/docs/reference/basic-syntax.html>
- ❑ <https://kotlinlang.org/docs/reference/coding-conventions.html>
- ❑ <https://kotlinlang.org/docs/reference/basic-types.html>
- ❑ <https://kotlinlang.org/docs/reference/classes.html>

- ❑ Android Documentation:
<https://developer.android.com/kotlin>